

Module IV

Queueing Theory

Queue or waiting lines are familiar phenomena which we observe quite frequently in our daily life.

For e.g., Customers entering a departmental store, Persons arriving at a Cinema Hall, Ships arriving at a Port, Truck arriving at a loading station, Letters arriving at a typist's desk, Patients arriving at a clinic, etc.

The objective of a queueing model is to find out the optimum service rate and the number of servers so that the average cost of being in a queueing system and the cost of service to be minimized.

Queueing theory is a mathematical discipline that deals with the study of queues (lines or waiting lines) and the behavior of entities in queues. It has wide-ranging applications in various fields where there are processes involving waiting and congestion. Some of the prominent areas where queueing theory is used include:

Telecommunications: Queueing theory is extensively used in the design and analysis of communication networks, such as telephone systems, data networks, and the internet. It helps in optimizing network performance, minimizing delays, and managing traffic congestion.

Transportation Systems: Queueing theory is applied to analyze and optimize traffic flow, transportation networks, and logistics systems. It helps in designing efficient transportation routes, scheduling services, and managing waiting times at transportation hubs like airports, bus stations, and train stations.

Manufacturing and Operations Management: Queueing theory is used in manufacturing processes to optimize production lines, inventory management, and supply chain operations. It helps in minimizing production delays, managing workloads, and improving overall efficiency.

Service Industry: Queueing theory plays a crucial role in service industries like healthcare (e.g., hospital emergency departments, appointment scheduling), retail (e.g., managing checkout lines), banking (e.g., ATM operations), and customer service centers. It assists in improving customer satisfaction, resource allocation, and service level agreements.

Computer Systems: Queueing theory is applied to the design and optimization of computer systems, such as operating systems, network protocols, and server farms. It helps in determining optimal system configurations to handle various workloads and prevent bottlenecks.

Call Centers: Call centers use queueing theory to manage call routing, agent staffing, and service level agreements. This helps in providing efficient customer support while minimizing waiting times.

Healthcare Systems: Queueing theory is used to model patient flow in hospitals, clinics, and healthcare facilities. It assists in optimizing appointment scheduling, resource allocation, and patient treatment processes.

Emergency Services: Queueing theory can be applied to emergency response systems, such as fire departments, police dispatch, and ambulance services. It aids in resource allocation and dispatching units effectively.

Public Services: Government agencies use queueing theory to manage services like passport offices, DMV (Department of Motor Vehicles) offices, and public welfare programs, ensuring efficient service delivery to citizens.

Research and Simulation: Queueing theory is also used in research and simulation studies across various disciplines to model real-world scenarios, predict system behavior, and test different strategies.

Queueing models are mathematical frameworks used to analyze and describe the behavior of queues or waiting lines in various systems. These models help in understanding the flow of entities (such as customers, data packets, or tasks) through a system where there is a process of waiting before they can be serviced or processed. Queueing models provide insights into factors such as waiting times, service rates, system utilization, and resource allocation. There are several types of queueing models, each with its own characteristics and assumptions. Some common queueing models include:

Single-Server Queue: In this model, there is one server that serves customers one at a time. The arrivals and service times may follow specific probability distributions. Examples include the M/M/1 (Poisson arrival, exponential service time) and M/D/1 (Poisson arrival, deterministic service time) models.

Multi-Server Queue: This model extends the single-server queue to have multiple servers. It is represented as M/M/c for c servers. Multi-server queues can handle higher arrival rates and provide faster service but are more complex to analyze.

Finite Queue Length Models: These models consider situations where there is a maximum limit on the number of entities that can wait in the queue. This is common in systems like call centers or customer service counters.

Priority Queue: In priority queues, different entities have different priority levels, and entities with higher priority are served before those with lower priority. Priority queue models are used in scenarios where certain entities need faster service than others.

Networks of Queues: This involves interconnected queues, often found in complex systems like computer networks. It models how entities move between different queues in a network.

Bulk Service Models: In some systems, multiple entities can be served together in a batch, which can affect waiting times and service rates. Bulk service models account for this behavior.

Queueing Networks: These models involve multiple queues that interact with each other, representing more complex systems like computer networks, manufacturing processes, and service centers.

Non-Exponential Models: While the exponential distribution is commonly used for arrivals and service times due to its memoryless property, real-world systems may have non-exponential distributions. Non-exponential models consider more realistic distributions.

Renewal Theory: Renewal theory deals with systems where arrivals are not governed by a continuous flow but occur at discrete intervals.

Markovian Models: Markovian queues consider the probabilistic behavior of entities based on the Markov process, where the future state depends only on the present state.

These models often involve parameters like arrival rates, service rates, queue lengths, and utilization rates. Queueing models can be analyzed using various techniques, such as probability theory, simulation, and numerical methods. They provide valuable insights for optimizing system performance, resource allocation, and decision-making in various fields.

The queuing system can be classified based on the following factors

i. The input or arrival pattern

This describes the manner in which the customers arrive and join the queueing system which is random. So the mode of arrival of customers is expressed by means of the probability distribution of the number of arrivals per unit time or of the interval time.

ii. The service mechanism or pattern

The service process should specify the number of servers and the arrangement of servers (in parallel or series) as behavior of the queue depends on this as well.

iii. The queue disciplines

This specifies the manner in which the customers from the queue are selected for the service.

FIFO/FCFS → First In First Out/First Come First Served

LIFO/LCFS → Last In First Out/Last Come First Served

SIRO → Service In Random Order

Kendall's Notation for representing Queueing Models

The queuing models are expressed as (A/S/C) : (K/D) Where

A → Arrival (or Interarrival) distribution,

S → Service Time (or departure) distribution

C → The number of service channels in the system.

K → Capacity of the system or Max number of units allowed in the system (service + waiting)

D → Service/Queueing Discipline

M → Markovian process

Note:

When the final two parameters are not specified it is assumed $K = \infty$ and $D = \text{FIFO}$

Here we deal with the service provided on FCFS or FIFO basis in Queueing system:

Symbolic representation of Queueing models:

1. Infinite capacity, single server(M/M/1: ∞/FIFO)
2. Infinite capacity, multiple server(M/M/S: ∞/FIFO)

Units in the System = Units in the Queue + Units that are being serviced

N- Number of customers in the queue system

(N-1 is the number of customers in the queue)

P_n - Probability that n number of customers arrive at any time t

W_s – The time between arrival and completion of service.

W_q – The time between arrival and reach of the service point

$E(N_s) = L_s$ - Average number of customers in the system

$E(N_q) = L_q$ – Average customers in the queue

$E(N_w) = L_w$ - Average number of customers in the non-empty queue

$E(W_s) = E(W)$ - Average waiting time of a customer in the system

$E(W_q)$ - Average waiting time of a customer in the queue

$f(w)$ - Probability density function of the waiting time in the system

Note:

1. Arrival rate follows the Poisson distribution
2. Service rate follows Exponential distribution
3. Service is provided on a first-come, first-served basis

Note:

- $\lambda \rightarrow$ Average arrival rate Average inter-arrival time is $1/\lambda$
- $\mu \rightarrow$ Average service rate $\Rightarrow$ Average service time is $1/\mu$

Characteristics of Infinite capacity, single server Poisson queue model I(M/M/1: ∞/FIFO model)
 $\lambda_n = \lambda$ & $\mu_n = \mu$

1. Probability that no customer is in the queue $= P_0 = 1 - \frac{\lambda}{\mu}$

Probability that n customers are in the queue $= P_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right)$

2. An Average number of customers in the system:

$$E(N_s) = L_s = \frac{\lambda}{\mu - \lambda}$$

3. Average number of customers in the queue or Average length of the queue

$$E(N_q) = L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

4. Average number of customers in the non-empty queue

$$E(N_w) = L_w = \frac{\mu}{\mu - \lambda}$$

5. Probability that the number of customers in the system exceeds k

$$P(N > k) = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

6. Probability that the waiting time of a customer in the system exceeds t

$$P(W_s > t) = e^{-(\mu - \lambda)t}$$

7. Probability density function of the waiting time in the system

$$f(w) = (\mu - \lambda)e^{-(\mu - \lambda)w}$$

8. Probability density function of the waiting time in the queue

$$g(w) = \begin{cases} \frac{\lambda}{\mu}(\mu - \lambda)e^{-(\mu - \lambda)w}, & w > 0 \\ 1 - \frac{\lambda}{\mu}, & w = 0 \end{cases}$$

9. Average waiting time of a customer in the system exceeds t

$$E(W_s) = \frac{1}{\mu - \lambda}$$

10. Average waiting time of a customer in the queue

$$E(W_q) = \frac{\lambda}{\mu(\mu - \lambda)}$$

11. Average waiting time of a customer in the queue, if he has to wait

$$E(W_q / W_q > 0) = \frac{1}{(\mu - \lambda)}$$

Little's Laws: These are the relations among $E(N_s)$, $E(N_q)$, $E(W_s)$, and $E(W_q)$ and hold good for the models with infinite capacity and with slight modification for the models with finite capacity.

i. $E(N_s) = \lambda E(W_s)$

ii. $E(N_q) = \lambda E(W_q)$

$$\text{iii.} \quad E(N_s) = E(N_q) + \frac{\lambda}{\mu}$$

iv. $E(W_s) = E(W_q) + \frac{1}{\mu}$

Problems:

- i. Arrival at a telephone booth is considered to be Poisson with an average time of 12 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponentially with mean 4 minutes. Find
- The average number of persons waiting in the system.
 - The probability that a person arriving at the booth will have to wait in the queue
 - Fraction of the day when the phone is in use.
 - The probability that it will take him more than 10min altogether to wait for the phone and complete his call
 - The average length of the queue that forms time to time.

Solution: Here, $\lambda = \frac{1}{12} / \text{minute}$ and $\mu = \frac{1}{4} / \text{minute}$

- The average number of persons waiting in the system.
 - The probability that a person arriving at the booth will have to wait in the queue
 - $P(\text{Phone is idle})$
- $P(\text{Phone will be in use}) =$

- d) The probability that it will take him more than 10min. altogether to wait for the phone and complete his call
 - e) The average length of the queue that forms time to time is the average length of the non empty queue
- ii. Customers arrive at a one-man barber shop according to a Poisson process with a mean inter-arrival time of 12 min. Customers spend an average of 10 minutes in the barber's chair.
- a) What is the expected number of customers in the barber shop and in the queue
 - b) Calculate the percentage of time an arrival can walk straight into the barber's chair without having to wait
 - c) How much time can a customer expect to spend in the barber's shop?
 - d) What is the average time customers spend in the queue
 - e) What is the probability that the waiting time in the system is greater than 30min.
 - f) Calculate the percentage of customers who have to wait prior to getting into the barber's chair
 - g) What is the probability that more than 3 customers are in the system
 - h) The owner will provide another chair and hire another barber, when a customer's waiting time in the shop exceeds 1.25hr. How much must be the average rate of arrivals increase to hire a second barber

Solution: Here, $\lambda = \frac{1}{12} / \text{minute}$ and $\mu = \frac{1}{10} / \text{minute}$

- a) The expected number of customers in the barber shop and in the queue

- b) Probability that the time an arrival can walk straight into the barber's chair without having to wait

The percentage of the time an arrival can walk straight into the barber's chair without having to wait =

- c) The time a customer expected to spend in the barber's shop=

- d) The average time customers spend in the queue=

- e) The probability that the waiting time in the system is greater than 30min

- f) Probability that a customer has to wait prior to getting into the barber's chair

The percentage of customers who have to wait prior to getting into the barber's chair

- g) What is the probability that more than 3 customers are in the system

- h) The owner will provide another chair and hire another barber, when a customer's waiting time in the shop exceeds 1.25hr.

The average rate of arrivals increase to hire a second barber=

- iii. At what average rate must a clerk in a supermarket work in order to ensure a probability of 0.90 that the customer will not wait longer than 12min?

It is assumed that there is only one counter at which customers arrive in a Poisson fashion at an average rate of 15 per hour and that the length of the service by the clerk has an exponential distribution.

Solution: Here, $\lambda = 15/ hr$

- iv. If people arrive to purchase movie tickets at the average rate of 6 per minute. It takes an average of 7.5 seconds to purchase a ticket. If a person arrives 2min before the movie starts and if it takes exactly 1.5min to reach the correct seat after purchasing the ticket.
- a) Can he expect to be seated for the start of the movie?
 - b) What is the probability that he will be seated for the start of the movie?
 - c) How early must he arrive in order to be 99% sure of being seated for the start of the movie?

Solution: Here, $\lambda = 6$ per minute and $\mu = 8$ per minute

a)

Average waiting time expected to be seated for the start of the picture=

Hence he can just be seated for the start of the picture.

a) Probability that he will be seated for the start of the picture=

b)

c)

- v. A duplicating machine maintained for office use is operated by an office assistant who earns Rs. 5 per hour. The time to complete each job varies according to an exponential distribution with mean 6min. Assume a Poisson input with an average arrival rate of 5 jobs per hour. If an 8-h day is used as a base, determine
- a) The percentage idle time of the machine
 - b) The average time a job is in the system and
 - c) The average earning per day of the assistant

Characteristics of Infinite capacity, multiple server Poisson queue model II(M/M/s:

∞/FIFO model) $\lambda_n = \lambda$ & $\mu_n = \begin{cases} n\mu & \text{if } 0 \leq n < s \\ s\mu & \text{if } n \geq s \end{cases}$

If there is single server, $\mu_n = \mu$ for all n but if there are s servers working independently of each other.

If the number of customers is less than s customers, $n < s$, only n of the s servers will be busy and the other servers idle. Hence the mean service rate will be $n\mu$.

If $n \geq s$, all s servers will be busy and hence the mean service rate = $s\mu$.

$$\text{i.e. } \mu_n = \begin{cases} n\mu & \text{if } 0 \leq n < s \\ s\mu & \text{if } n \geq s \end{cases}$$

Fraction of the time the a server is busy = Traffic intensity = $\frac{\lambda}{\mu s}$

$\frac{\lambda}{\mu s} < 1$ in order that infinite queue is not formed in multiserver model.

Note: $\lambda < \mu s$

1.

$$P_0 = \frac{1}{\left\{ \sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n \right\} + \left\{ \frac{1}{s! \left(1 - \frac{\lambda}{\mu s} \right)} \left(\frac{\lambda}{\mu} \right)^s \right\}}$$

$$P_n = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n P_0, \quad \text{if } 0 \leq n < s$$

$$P_n = \frac{1}{s! s^{n-1}} \left(\frac{\lambda}{\mu} \right)^n P_0, \quad \text{if } n \geq s$$

2. Average number of customers in the system:

$$E(N_s) = L_s = \frac{\left(\frac{\lambda}{\mu} \right)^{s+1}}{s \cdot s! \left(1 - \frac{\lambda}{\mu s} \right)^2} P_0 + \frac{\lambda}{\mu}$$

3. Average number of customers in the queue or Average length of the queue

$$E(N_q) = L_q = \frac{\left(\frac{\lambda}{\mu} \right)^{s+1}}{s \cdot s! \left(1 - \frac{\lambda}{\mu s} \right)^2} P_0$$

4. Average waiting time of a customer in the system

$$E(W_s) = \frac{1}{\mu} + \frac{\left(\frac{\lambda}{\mu} \right)^s}{\mu \cdot s \cdot s! \left(1 - \frac{\lambda}{\mu s} \right)^2} P_0$$

5. Average waiting time of a customer in the queue

$$E(W_q) = \frac{\left(\frac{\lambda}{\mu}\right)^s}{\mu \cdot s \cdot s! \left(1 - \frac{\lambda}{\mu s}\right)^2} P_0$$

- 6. Probability that an arrival has to wait or probability that there are s or more customers in the system**

$$P(W_s > 0) = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s! \left(1 - \frac{\lambda}{\mu s}\right)}$$

- 7. Probability that an arrival enters the service without waiting**

$$= 1 - \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s! \left(1 - \frac{\lambda}{\mu s}\right)}$$

- 8. Mean waiting time in the queue for those who actually wait**

$$E(W_q/W_q > 0) = \frac{1}{\mu s - \lambda}$$

- 9. Probability that there will be someone waiting**

$$P(N \geq s + 1) = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0 \left(\frac{\lambda}{\mu s}\right)}{s! \left(1 - \frac{\lambda}{\mu s}\right)}$$

- 10. Average number of customers in non empty queue**

$$E(N_w) = L_w = \frac{\left(\frac{\lambda}{\mu s}\right)}{\left(1 - \frac{\lambda}{\mu s}\right)}$$

1. There are three typists in an office. Each typist can type an average of 6 letters per hour. If letters arrive for typing at the rate of 15 letters per hour.

- a) What is the probability that all the typists will be busy
- b) What is the average number of letters waiting to be typed
- c) What is the average time a letter has to spend waiting and being typed

Solution: Here, $\lambda = 15$ *per hour* and $\mu = 6$ *per hour*

a)

The probability that all the typists will be busy=

b) The average number of letters waiting to be typed

c) The average time a letter has to spend for waiting and for being typed

2. Given an average arrival rate of 20 per hour, is it better for a customer to get service at a single channel with mean service rate of 22 customers per hour at one of two channels in parallel with mean service rate of 11 customers per hour for each of the two channels. Assuming both the channels to be Poisson type.

Solution: Here, $\lambda = 20$ per hour and $\mu = 22$ per hour, $s=1$

$$\lambda = 20 \text{ per hour and } \mu = 22 \text{ per hour, } s=2$$

3. In a supermarket, two girls are attending to customers at the counters. If the service time for each customer is exponential with a mean of 4 minutes, and if people arrive in Poisson fashion at the rate of 10 per hour.

- a) What is the Probability that a customer has to wait for service?
- b) What is the expected percentage of idle time for each girl?
- c) If the customer has to wait in the queue, what is the expected length of his waiting time

Solution: $\lambda = 10$ per hour $= \frac{1}{6}$ min and $\mu = 22$ per hour $= \frac{1}{4}$ per min, $s=2$

- a) Probability that a customer has to wait for service

b) Fraction of time when the girl is busy

Fraction of time when the girl is idle=

The expected percentage of idle time for each girl=

c) If the customer has to wait in the queue, the expected length of his waiting time=

4. A petrol pump station has 4 pumps. The service times follow the exponential distribution with a mean of 6min and cars arrive for service in a Poisson process at the rate of 30 cars per hour.

- a) What is the probability that an arrival would have to wait in line
- b) Find the average waiting time, average time spent in the system and the average number of cars in the system
- c) For what percentage of time would a pump be idle on an average